

Real Numbers (Set B) — MCQ Worksheet

Mathematics • Chapter 1 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The number of prime factors in the factorisation of 8820 is:

- | | |
|-------|-------|
| (A) 3 | (B) 4 |
| (C) 5 | (D) 6 |

Q2. HCF of 96 and 120 is:

- | | |
|--------|--------|
| (A) 12 | (B) 24 |
| (C) 48 | (D) 8 |

Q3. LCM of 12 and 18 is:

- | | |
|--------|---------|
| (A) 24 | (B) 36 |
| (C) 72 | (D) 108 |

Q4. Which of these is irrational?

- | | |
|-----------------|---------------------|
| (A) $\sqrt{25}$ | (B) $\sqrt[3]{16}$ |
| (C) $\sqrt{7}$ | (D) $\sqrt[3]{100}$ |

Q5. The decimal expansion of $7/8$ is:

- | | |
|-----------------------------------|-------------------------------|
| (A) Terminating | (B) Non-terminating recurring |
| (C) Non-terminating non-recurring | (D) Irrational |

Q6. If $\text{HCF}(a, b) = 12$ and $a \times b = 1800$, then $\text{LCM}(a, b)$ is:

- | | |
|---------|---------|
| (A) 150 | (B) 120 |
| (C) 180 | (D) 100 |

Q7. The smallest number divisible by both 8 and 12 is:

- | | |
|--------|--------|
| (A) 20 | (B) 24 |
| (C) 48 | (D) 96 |

Q8. Sum of a rational and an irrational number is always:

- | | |
|--------------|----------------|
| (A) Rational | (B) Irrational |
| (C) Integer | (D) Zero |

Q9. The largest number that divides 70 and 125 leaving remainders 5 and 8 respectively is:

- | | |
|---------|----------|
| (A) 13 | (B) 65 |
| (C) 875 | (D) 1750 |

Q10. If p is prime, then $\sqrt{p}$ is:

- | | |
|----------------|----------------|
| (A) Rational | (B) Irrational |
| (C) An integer | (D) Zero |

Q11. Which of these has terminating decimal expansion?

- | | |
|--------------|--------------|
| (A) $1/7$ | (B) $3/14$ |
| (C) $29/343$ | (D) $17/200$ |

Q12. The prime factorisation of 1024 is:

- | | |
|---------------------|--------------------|
| (A) 2^{10} | (B) $2^8 \times 4$ |
| (C) $2^5 \times 32$ | (D) 4×256 |

Q13. HCF (135, 225) by Euclid's algorithm is:

- | | |
|--------|--------|
| (A) 15 | (B) 25 |
| (C) 45 | (D) 75 |

Q14. Which of the following is a composite number?

- | | |
|--------|--------|
| (A) 11 | (B) 13 |
| (C) 21 | (D) 17 |

Q15. If two positive integers a and b can be written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime, then HCF(a, b) is:

- (A) xy^2 (B) x^3y^3
(C) x^3y^2 (D) xy

Q16. Three bells toll at intervals of 9, 12 and 15 minutes. If they start tolling together, after how many minutes will they next toll together?

- (A) 120 min (B) 180 min
(C) 60 min (D) 30 min

Q17. The least number that when divided by 12, 16, 24 and 36 leaves remainder 7 in each case is:

- (A) 151 (B) 144
(C) 137 (D) 127

Q18. Show that $4n$ cannot end with digit 0 for any natural number n because:

- (A) $4n$ has only 2 as its prime factor (B) $4n$ is always even
(C) $4n$ is a perfect square (D) $4n$ is divisible by 10

Q19. Two numbers are in ratio 15:11. If their HCF is 13, the numbers are:

- (A) 195, 143 (B) 75, 55
(C) 150, 110 (D) 165, 121

Q20. The HCF and LCM of two numbers are 33 and 264 respectively. If one number is 132, the other is:

- (A) 66 (B) 264
(C) 33 (D) 198

Q21. If $5 + 2\sqrt{3} = a + b\sqrt{3}$ where a, b are rational, then:

- (A) $a = 5, b = 2$ (B) $a = 2, b = 5$
(C) $a = 7, b = 0$ (D) $a = 0, b = 7$

Q22. Express $0.5\overline{5}$ (i.e. $0.555\dots$) as a fraction:

- (A) $1/2$ (B) $5/9$
(C) $5/10$ (D) $1/5$

Q23. A merchant has 120 litres of oil of one kind, 180 litres of another and 240 litres of a third kind. He wants to sell oil by filling the three kinds in tins of equal capacity. Largest such capacity is:

- (A) 60 L (B) 30 L
(C) 40 L (D) 80 L

Q24. Find the largest number that divides 245 and 1029 leaving remainder 5 in each case.

- (A) 16 (B) 12
(C) 8 (D) 24

Q25. Without actual division, decide whether $64/(2^5 \times 5^2)$ has terminating decimal expansion:

- (A) Yes, terminating (B) No, non-terminating
(C) Cannot determine (D) Equals zero